

Practice Guidance

Critical Update: AASLD Practice Guidance on prevention, diagnosis, and treatment of hepatocellular carcinoma

BACKGROUND

The purpose of this brief update is to ensure awareness of a change in the AASLD HCC guidance recommendations^[1] based on newly published clinical trial results that impact clinical practice. We provide revision to selected text, guidance statement 32, and [Figure 11](#) in response to new data from an updated analysis of the IMbrave050 study, a randomized multicenter trial of adjuvant atezolizumab + bevacizumab after resection/ablation therapy in patients at high risk of recurrence versus active surveillance. ^[2] In this updated analysis, the originally reported^[3] benefit in the primary endpoint of recurrence-free survival with atezolizumab + bevacizumab was not sustained. Similar to the first interim analysis, overall survival remained non-significant at the second interim analysis. Based on these updated data, the risk/benefit ratio does not support the use of atezolizumab + bevacizumab as adjuvant therapy for high-risk HCC after resection/ablation. Publishing guidance in a rapidly advancing field is challenging. AASLD is considering innovative methods to bring timely updates, such as this one, to the field.

UPDATED GUIDANCE ON IMMUNOTHERAPY IN THE ADJUVANT SETTING

The following revision updates the text in the original guidance document,^[1]

beginning with the second paragraph under the section entitled, “Risk of HCC recurrence and use of (neo)adjuvant therapy.” Preserved text is in quotations (apart from renumbered references), and altered text is underlined.

“There is a need for effective (neo)adjuvant therapy to reduce risk of HCC recurrence after surgical resection. In HCV-associated HCC, two large multicenter studies from North America^[4] and Italy^[5] confirmed that eradication of HCV with direct-acting antiviral therapy does not increase risk of HCC recurrence and improves survival. Preoperative TACE in patients with large resectable HCC does not improve recurrence-free survival and may increase risk of interval tumor progression, precluding surgical resectability.”^[6] Current data do not support the use of neoadjuvant systemic therapies in patients with HCC undergoing surgical resection.

“An RCT of adjuvant sorafenib in patients with HCC undergoing resection or thermal ablation did not improve recurrence-free survival in patients compared with placebo (HR = 0.94; 95% CI: 0.78–1.13).^[7] The open-label phase III RCT comparing atezolizumab plus bevacizumab versus active surveillance (IMbrave050) in the adjuvant setting for HCC patients at high-risk of recurrence after resection or local ablation” initially demonstrated positive results.^[3] “After a median follow-up of 17.4 months, the trial hit its

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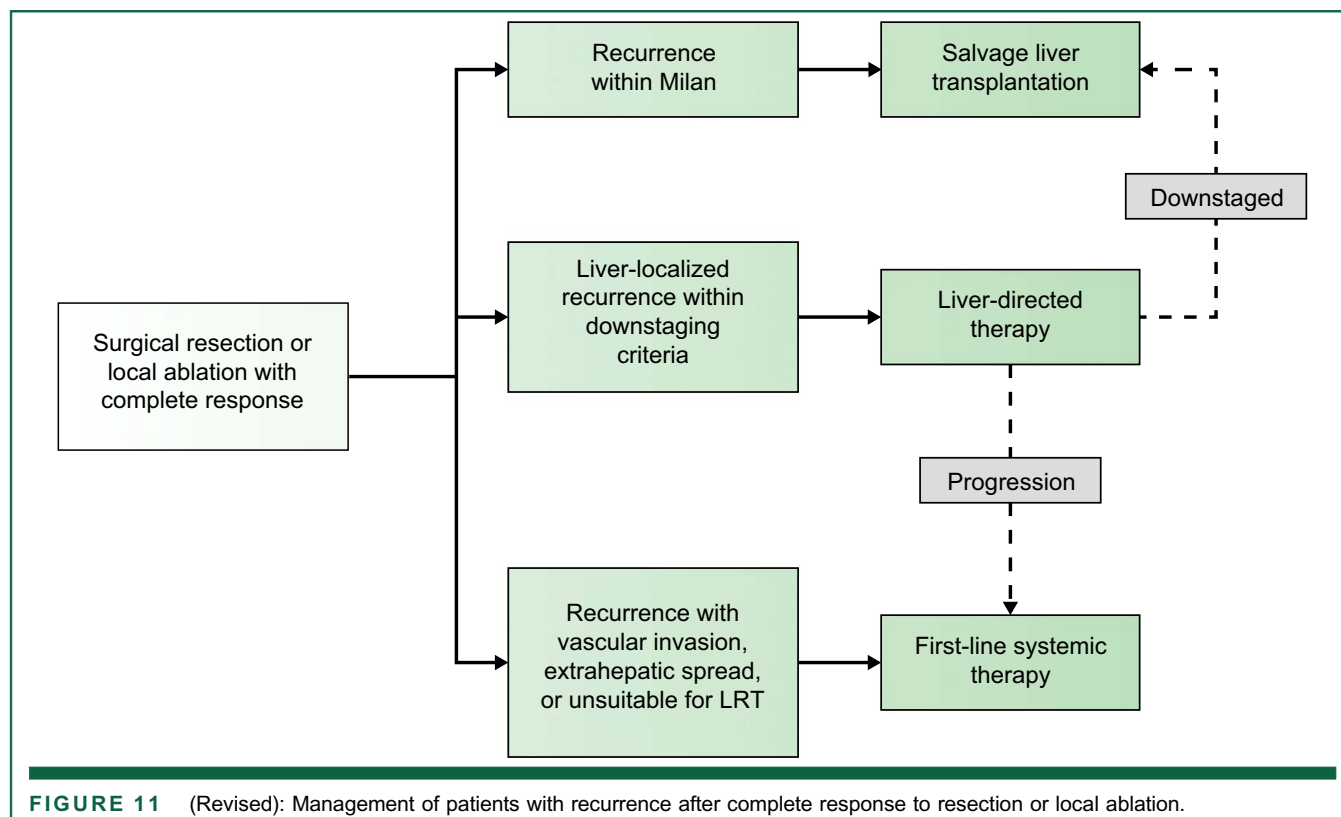
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primary endpoint for superiority of recurrence-free survival (RFS = 0.72, 95% CI: 0.56–0.93), with 12-month recurrence-free survival estimates of 78% versus 65% for the intervention and surveillance arms, respectively. The median duration of treatment for atezolizumab plus bevacizumab was 11 months, with 34.9% of patients experiencing

Abbreviation: RCT, randomized controlled trial.

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grade 3-4 treatment related adverse events.” However, in a second interim analysis after a median follow-up of 35.1 months, the recurrence-free survival benefit observed in patients treated with atezolizumab+bevacizumab versus active surveillance at the primary analysis was not sustained over time (HR=0.90; 95% CI: 0.72–1.12).^[2] Overall survival continued to be immature at the updated analysis, with >80% of patients alive in both arms after 2 years (HR=1.26; 95% CI: 0.85–1.87). Across subgroups, recurrence-free survival and overall survival results were consistent. No new safety concerns were noted. Continued follow-up is ongoing.

The optimal management of patients with recurrence after margin-free resection or complete response following local ablation should follow the criteria established for evidence-based management according to distinct stages described in this guidance update, as summarized in [Figure 11](#) (Revised). Salvage transplantation has been recommended for eligible patients with postsurgical recurrence within Milan Criteria. Patients with liver-localized recurrence beyond Milan Criteria can be treated with liver-directed therapy, with consideration of liver transplantation in those who are successfully downstaged. “Patients with vascular invasion, extrahepatic spread, or TACE unsuitable disease should be considered for systemic therapy.”

“Recent proof-of-principle studies utilizing neoadjuvant systemic therapies prior to surgical resection have been reported, and there are several ongoing adjuvant and neoadjuvant phase II–III RCTs. In a single-arm phase Ib study, neoadjuvant cabozantinib plus nivolumab in 15

patients with borderline resectable HCC allowed for margin-negative resection in 80% of patients, with 42% having a major pathologic response.^[8] In another single-center phase II study of neoadjuvant nivolumab with or without ipilimumab, 6 (30%) of 20 patients who underwent resection had a major pathologic response.”^[9] *AASLD advises against the use of adjuvant or neoadjuvant systemic therapies in patients undergoing liver resection.*

Guidance statement

32 (Revised). AASLD advises against the use of adjuvant and neoadjuvant systemic therapies in patients undergoing liver resection/local ablation (**Level 1, Strong Recommendation**).

CONTEXT

Over the last 2 decades, recurrence-free survival has been accepted by scientific societies and regulatory bodies as the primary endpoint in randomized controlled trials (RCTs) assessing the clinical benefit of adjuvant therapy after resection or local ablation of HCC.^[10] This composite endpoint combines 2 events: recurrence, which is a major complication in 40% of patients at 3 years, or death, which may result as a complication of cirrhosis or as a treatment-related adverse event. While IMbrave050 was the first RCT in the adjuvant arena to show significant benefit in

recurrence-free survival at the first interim analysis, on subsequent analysis, this benefit was no longer demonstrated. This trial provides a cautionary note to assessing data at early interim time points and underscores the importance of designing an interim analysis that is robust enough to call a trial positive as per conventional stopping rules. The reason for the initially positive and ultimately negative recurrence-free survival results may be related to the lack of proportion of hazards, which can affect the outcome and magnitude of benefit.^[11] The issue of nonproportional hazards in the context of halting trials at early interim analysis, particularly in the context of neoadjuvant and adjuvant studies in HCC, will require a thorough analysis. In this regard, a phase III study in triple-negative breast cancer has already been reported assessing restricted mean survival time because the proportional-hazards assumption was not valid.^[12] Secondary endpoints recommended in the adjuvant setting include overall survival and patient-reported outcomes.

Given the final results of IMbrave050, surveillance remains the current standard of care for management after resection/local ablation with the intention of cure. Several phase III trials of single agent or combination therapies versus surveillance are currently ongoing in the adjuvant space, including pembrolizumab (KEYNOTE-937), nivolumab (CheckMate-9DX), durvalumab plus bevacizumab (EMERALD-2), or camrelizumab plus apatinib. In the neoadjuvant setting, for which pathological response at resection and recurrence-free survival is considered adequate endpoints, phase II studies are exploring combinations such as atezolizumab+bevacizumab, nivolumab+ipilimumab, or lenvatinib+pembrolizumab.^[13]

CONCLUSIONS

There are currently no FDA-approved (neo)adjuvant therapies for HCC. Surveillance for recurrence is the current standard of care after resection/local ablation with the intention of cure, even in patients with a high risk of recurrence. (Neo)adjuvant therapies should be considered only in the context of a clinical trial.

CONFLICTS OF INTEREST

Mark Yarrowan consults for Exelixis, Genentech, and AstraZeneca. Josep M. Llovet consults for Eisai, Merck, Roche, Genentech, AstraZeneca, Bayer, AbbVie, Sanofi, Moderna, Glycotest, and Exelixis. He is on the data safety monitoring board for BMS. All associated fees are paid to his institution. The remaining authors have no conflicts to report.

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